



SIMATS ENGINEERING

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Course Code: CSA1407

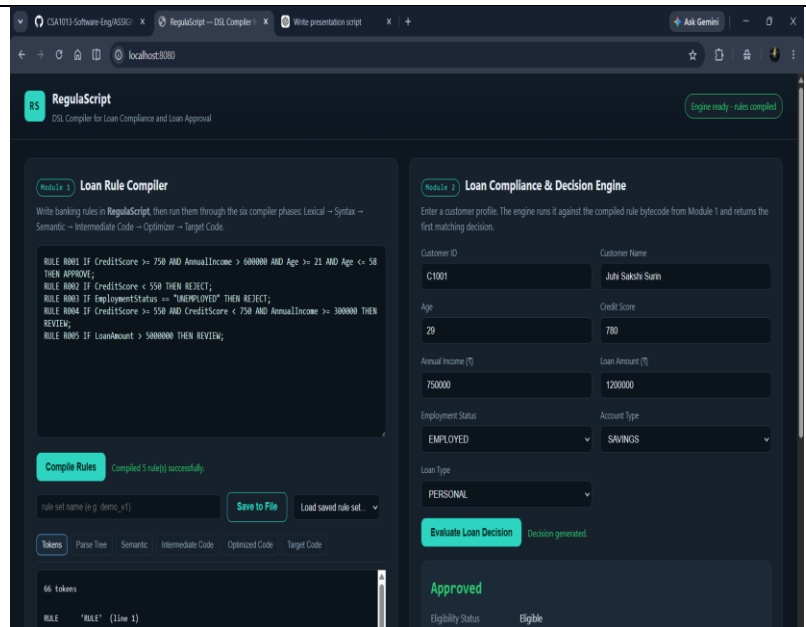
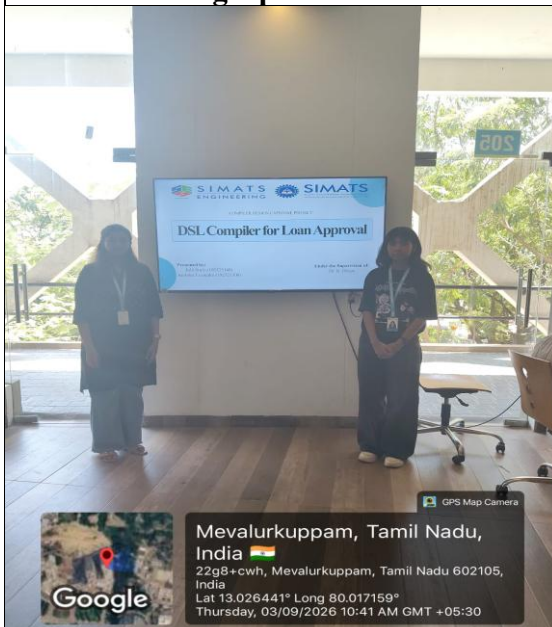
Slot: D

Course Name: Compiler Design

Course Faculty: Dr. N. Deepa

Project Title: DSL Compiler for Loan Approval

Module Photographs:



Project Description: Module 1: Loan Rule Compiler

The project, **DSL Compiler for Loan Approval**, introduces **RegulaScript**, a simple Domain-Specific Language designed for defining loan approval rules. Instead of manually coding banking rules, users can specify them using RegulaScript. These rules are processed through different compiler phases, including lexical analysis, syntax analysis, semantic analysis, and optimization, and are then converted into executable logic. The system evaluates customer information based on these rules and generates a loan eligibility decision, making the approval process more consistent, transparent, and less prone to manual errors.

The **Loan Rule Compiler** is the first module of the system. It accepts loan approval rules written in RegulaScript as input and processes them through the compiler pipeline. Lexical analysis identifies tokens such as keywords, identifiers, operators, and literals, while syntax analysis checks whether the rules follow the correct grammar. Semantic analysis validates the field names, data types, and logical conditions. After processing and optimization, the rules are converted into executable logic and passed to the next module for evaluating customer loan eligibility.

Student Signature

Guide Signature